Statistical Tests For Means

An e-commerce company claims that itys average delivery time is 30 minutes. We collect data from 50 deliveries and find an average of 32 minutes.. Can we conclude that the company's claim is wrong?
Answer: Not immidiately. The difference could be due to random variation in th sample.

Technically:
A statistical hypothesis test uses sample data to evaluate a claim about a population parameter

Why?
1. Compare actual performance against the target.
2. compare sales between two stores.
3. Measure the impact of employee training
4. Evaluate whether a new process reduces delivery time.

Z- Test: When and Why

A **Z-Test** is a statistical test that uses the standard normal distribution to evaluate a hypothesis about a population mean or a difference between means.

For a one-sample mean Z-test, the popualtion standard deviation must be known.

Xbar = Sample mean

$$Z = \frac{\bar{x} - \mu_0}{\underbrace{\sigma/\sqrt{n}}_{SE}}$$

Important A large sample does not automatically make every Z-Test appropriate.The test and its assumptions must match the problem.

One Sample Z test

The one sample Z-test compares a sample mean against a specified population mean when the population satandard deviation is know.

Delivery time:

Claimed mean	30
Sample Mean	32
Population SD	6
Sample Size	36
Significance level	0.05

Step 1: Formualate the hypothesis

H0:μ=30

H1:μ≠30

Step 2: Calculate the SE

SE = σ/SQRT(n)

SE = 6/SQRT(36) 1

Step 3: Calculate the Z statistic

Z = (32-30)/1

Z = 2

Step 4: Find the P value

p = 2*[1-ϕ(|Z|)]

For Z = 2 0.0228 0.0456

p≈0.0456 0.9772

0.045500264

Step 5: Make the decision

Since 0.0456<0.05

Decision: Reject H0

There is statistically significant evidence that the population mean differs ffrom 30 minutes.

Two Sample Z-Test

A two sample Z-test compares the means of two independent populations when their population Standard deviations are known.

Example:Marketing Campaign

A company wants to determine whether two campaigns produce different average sales.

Parameter	Campaign A	Campaign B
Sample Mean	₹ 500.00	₹ 540.00
Population SD	₹ 100.00	₹ 120.00
Sample Size	100	100

Step 1: State the hypothesis

H0:μA−μB=0

H1:μA−μB≠0

Step 2: Calculate the Standard Error

$$SE = \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}$$

SQRT((100^2/100)+(120^2/100)) = SQRT(100+144)

SE = 15.62049935

Step: 3 Calculate the Z- Statistic

Z = (Sample Mean A -Sample Mean B)/ SE

=(500-540)/15.62

-2.560819462

Z≈−2.56

0.0052 0.0104

The two tailed p- value is approximately 0.0104

since p < 0.05, reject H0.

Busineness Interpretion

The data provides statistically significant evidence that the population mean sales differ betweeen the campaigns

However, the test alone does not estasblish why the difference occurred or whther the additional campaign costs are justified.



Difference between one-sample Z-test and two sample Z- test

One sample Z-test we compare the Sample mean with the specified population mean

Two Sample Z- test we compare the Sample mean of Two indpendent populations

<u>Assumptions of Z- Test</u>

1. Population SD is known
2. Random Sampling
3. Indpendence
4. Normality or a suitable large- sample approximation
5. Independent groups for two sample test.

Always check the data structure before chossing the test.

T- Test: When and Why?

A **T-test** is a statistical test used to evaluate hypothesis about population means when the population Standard deviation is unknown and must be estimated from the sample data.

Why?

In real business situations, the population standard deviaition is rarely known.

A test uses the sample standard deviation to estimate the uncertainty

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

s= Sample SD

n = Sample Size

df = n-1

Z test vs T test	Z - Test	T-Test
Population SD	Known	Unknown
Variability used	Population SD	Sample SD
Reference Distribution	Standard Nrml	t-distribution

Important:

Do not select Z-test or T-test based only on whther n is above 30 or belove 30.

Whether the populaiton SD is known, the data structure and the test assumption matter.

One sample T-Test

A one sample T-test compares a sample mean with a specified population mean when the population standard deviation is unknown.

Business Example: Call center performance

A call center claims its average handling time is 5 minutes.

Hypothesized Mean	5 minutes
Sample Mean	5.2 minutes
Sample SD	0.2 minutes
Sample Size	10

Step 1: Formulate the Hypothesis

H0:$\mu=5$

H1:$\mu\neq5$

Step 2: Calculate the SE

SE = s / SQRT(n) = 0.2/sqrt(10)

0.063245553

SE≈0.0632

Step 3: t- statistic

t = (5.2-5)/0.0632

3.16227766

t≈3.16

Degree of freedom:

df = 10-1 = 9

	0.98	0.99
	2.821	3.25
	0.02	0.01
		0.0114
At ∝ = 0.05, reject H0.		0.01155

There is statistically significant evidence that the average call handling time differs from 5 min.

Two Sample T- Test.

An indepnedent two sample T-test compares the population means of two independent groups.

	A	B	
Sample Size	25	25	
Avg task	28	32	
Sample SD	5	6	H0:μA=μB
			H1:μA≠μB

Unequal Variance(Standard Deviaiton)--> Welch's T-test

0.882761246

If p < 0.05, reject H0

If p ≥ 0.05, fail to reject H0

Paired Sample T- Test

A paired sample T-test compares two related measurements, such as observations taken from the same people before and after intervention.

Business Example: Training Effectiveness

Measure the porductivity of same employees before and after the training

Employee	Before	After	Difference
A	20	25	5
B	24	27	3
C	22	26	4
D	18	22	4
E	26	29	3

test whether the populaiton mean difference is zero

t = d/sd/sqrt(n)

d= mean of paired differences
sd = SD of paired differences
n = number of pairs

0.000529311

The test evaluates whther the average before and after change differs from zero.

Different impoyees iun two groups --> independent t- test
sample employees measures twice --> Paiered T- test.